

Twitter User Timeline Analytics & Engagement Prediction System

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Abstract:

Social media platforms such as Twitter generate millions of user-generated posts every day, representing valuable insights into user behavior, sentiment trends, and engagement patterns. This project focuses on analyzing Twitter user timelines using a cleaned dataset of tweets. The system performs sentiment analysis, engagement trend evaluation, tweet activity tracking, and linguistic pattern visualization. Key tasks include data preprocessing, feature analysis, and generating visual insights using Python-based tools. This documentation provides an overview of the project's objectives, methodology, functional requirements, system architecture, and instructions for running the application.

Introduction:

Twitter (now known as X) is a high-velocity social platform where millions of short messages are shared every day, offering a rich source of information about how people communicate, react, and engage online. This project builds a complete analytical pipeline on top of a cleaned timeline dataset to study user behavior and tweet performance over time. It combines sentiment analysis, engagement modelling (based on likes, replies, retweets, and other interactions), and temporal activity analysis to uncover which types of content and posting patterns drive higher visibility. Using Python-based processing, feature extraction, and visualization, the system transforms raw tweet streams into interpretable insights for marketers, influencers, brands, and researchers, and extend the analysis.

Problem Statement:

Twitter users generate millions of tweets daily, yet it remains unclear which factors determine the level of engagement a tweet receives. Engagement can vary based on:

- Time of posting
- Tweet length
- Number of hashtags
- User popularity
- Tweet content and metadata

The challenge is to analyze these factors, build predictive models, and generate visual insights

Dataset Description:

Feature	Description
UserID	Unique user identifier
TweetID	Unique tweet identifier
timestamp	Date & time of tweet
Likes	Number of likes
RetweetCount	Number of retweets
text_len	Length of tweet text
num_hashtags	Count of hashtags
num_mentions	Number of @mentions
num_urls	URLs included
day_name	Day of week
hour	Hour of posting
engagement_score	Combined score = likes + retweets + replies

The dataset is rich enough to analyze both user-level and tweet-level behavior.

Objectives:

This project aims to:

(1) Analyze user behavior and tweet patterns

- Frequency of tweets
- Best posting hours
- Day-wise tweet trends

- Top users by activity and engagement

(2) Build predictive models for engagement

Models used:

- XGBoost Regressor
- Linear Regression (StatsModels OLS)
- Prophet Time-Series Forecasting
- Deep Learning Model (PyTorch)

(3) Generate interactive & static visualizations

- Tweet frequency per hour
- Engagement distribution
- Correlation heatmap
- Likes vs retweets
- User timeline trends
- Top users and top tweets

Tools & Technologies:

Category	Tools Used
Programming Language	Python
Data Manipulation	Pandas, NumPy
Visualization	Matplotlib, Seaborn, Plotly
Machine Learning	XGBoost, Scikit-learn
Deep Learning	PyTorch
Statistical Modeling	StatsModels
Time-Series	Prophet
IDE	Jupyter Notebook / VS

Functional Requirements:

- Load raw Twitter dataset
- Clean and preprocess data
- Engineer features (textual, temporal, user aggregates)
- Calculate engagement scores
- Analyze user-level and tweet-level behaviors
- Identify best posting hour per user

Methodology:

Step 1 — Data Cleaning

- Removed missing values
- Fixed timestamp formats
- Cleaned user IDs and tweet IDs
- Standardized engagement score metrics

Step 2 — Feature Engineering

The following derived features were added:

Feature	Purpose
text_len	Length of tweet to test if longer tweets perform better
sin_hour, cos_hour	Time-cycle encoding
Likes_mean	User's historical average likes
RetweetCount_mean	User's historical average retweets
Hour & Day columns	Time-based behavioral patterns

Step 3 — Predictive Modeling

XGBoost Regressor Results

- MSE = 472.48

- $R^2 = 0.7766$ (good performance)

StatsModels Linear Regression

- $R^2 = 0.852$ (excellent linear fit)
- Tweet length, likes_mean, retweet_mean = strongest predictors.

Prophet Forecasting

Performed time-series forecasting of engagement over the hours of the day.

PyTorch Neural Network

Trained a deep-learning engagement predictor.

Basic Instructions to Run the Application:

Step 1: Install Required Libraries

`pip install pandas`

`pip install numpy`

`pip install matplotlib`

`pip install seaborn`

`pip install wordcloud`

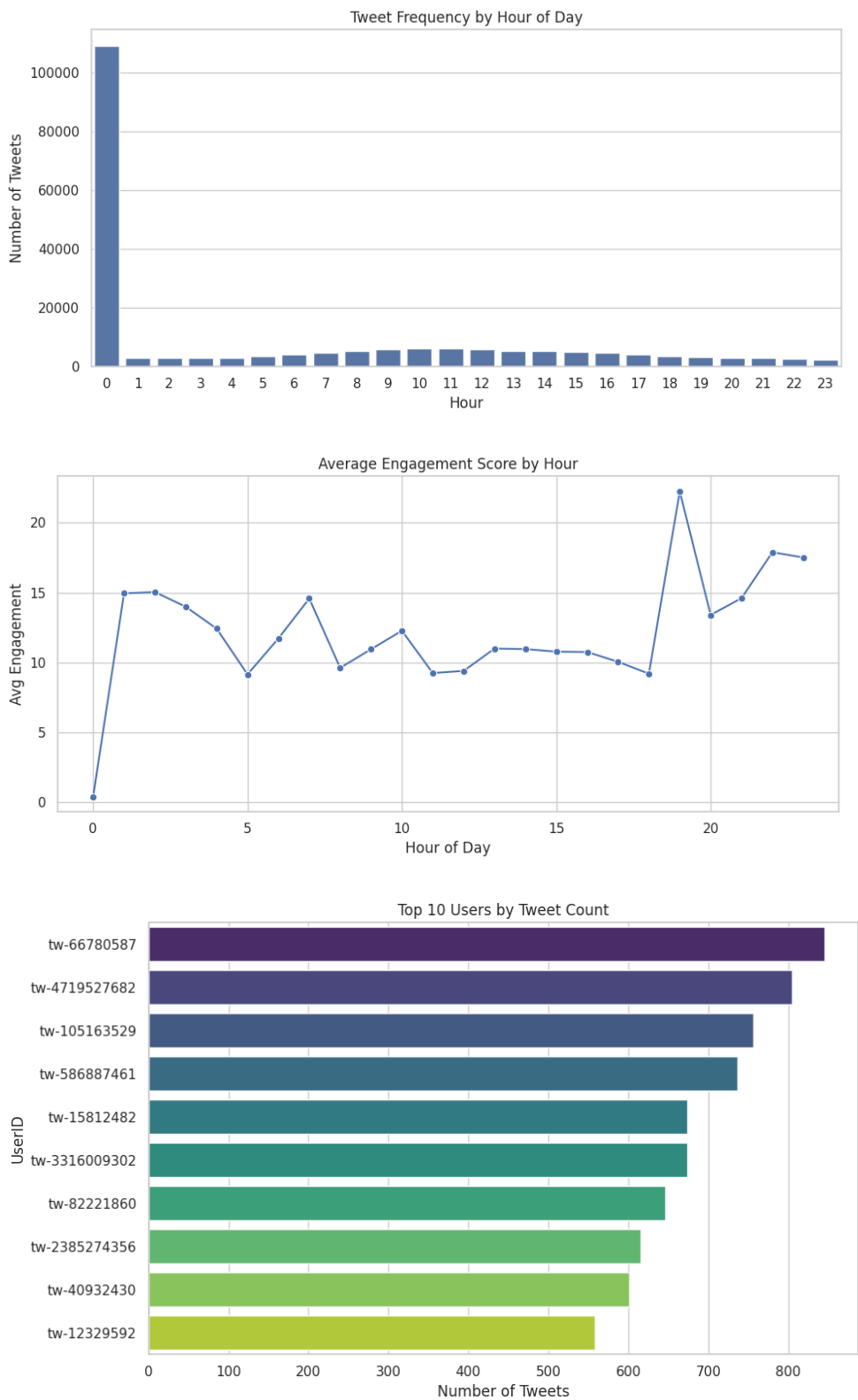
`pip install textblob`

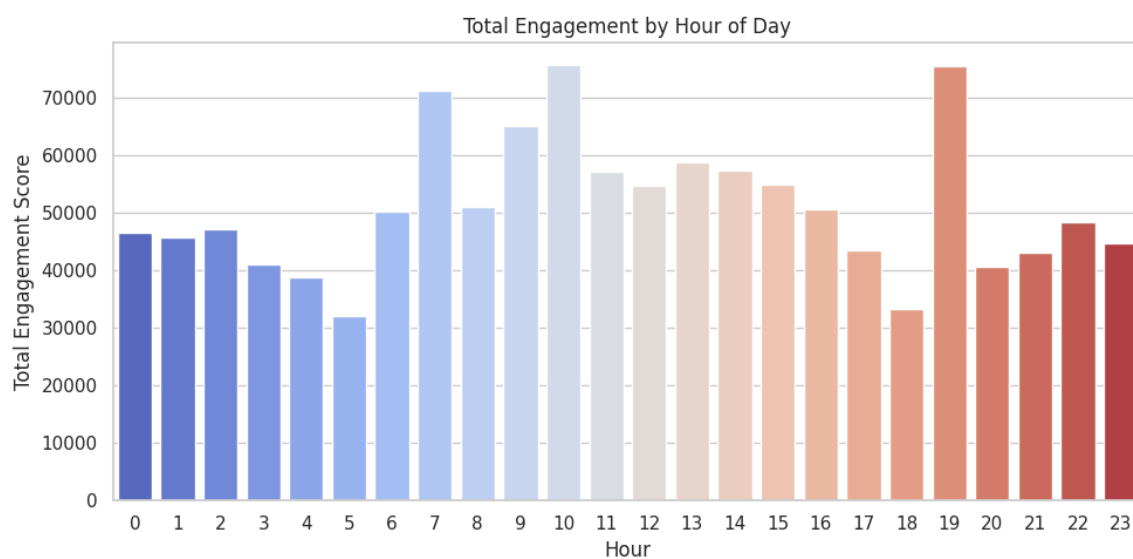
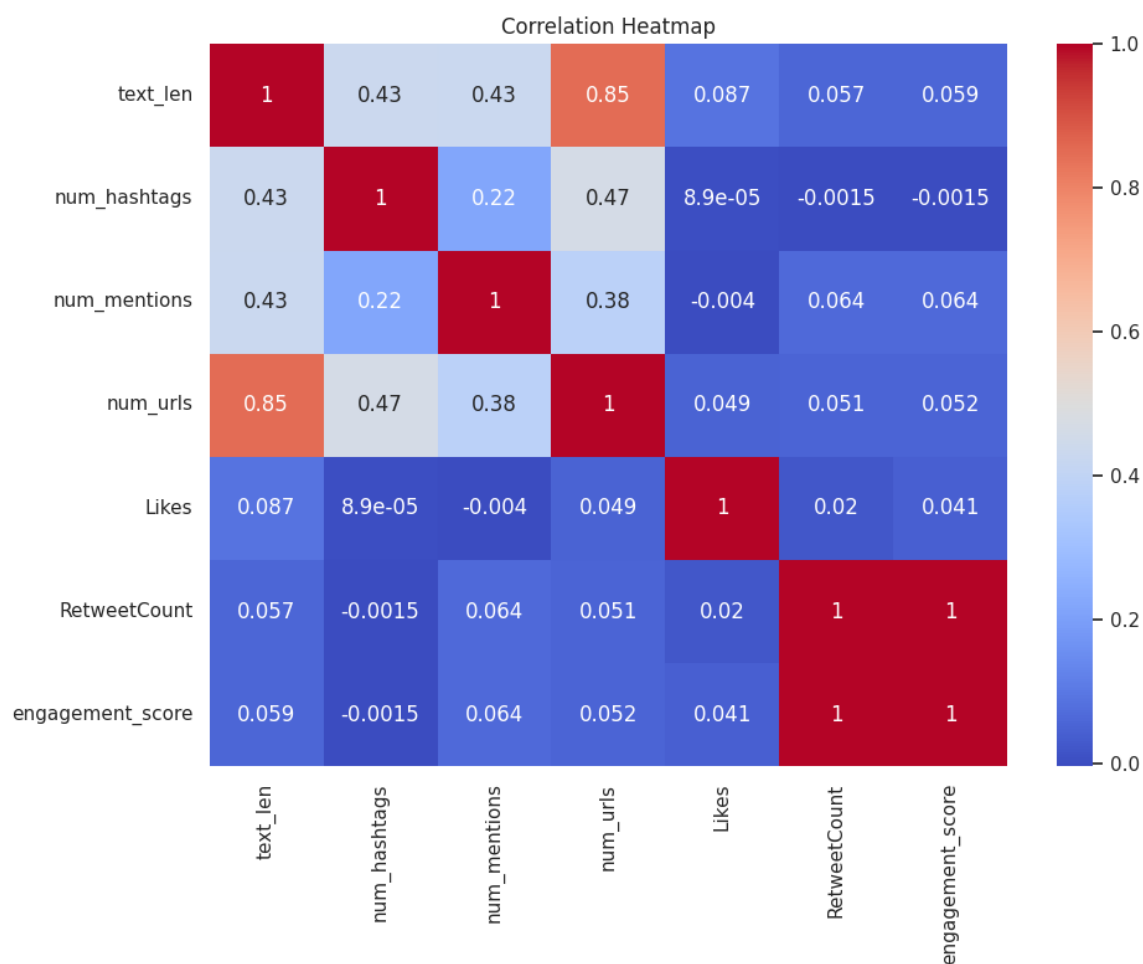
Step 2: Place Dataset

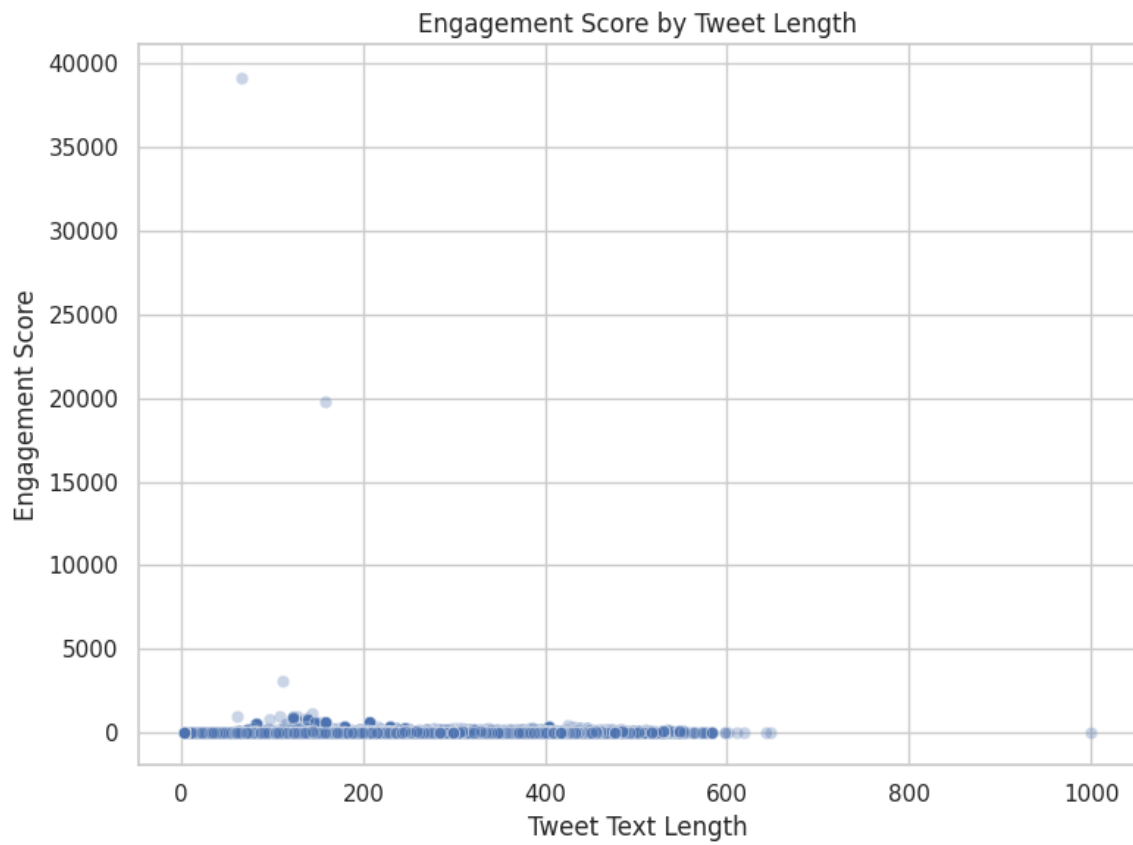
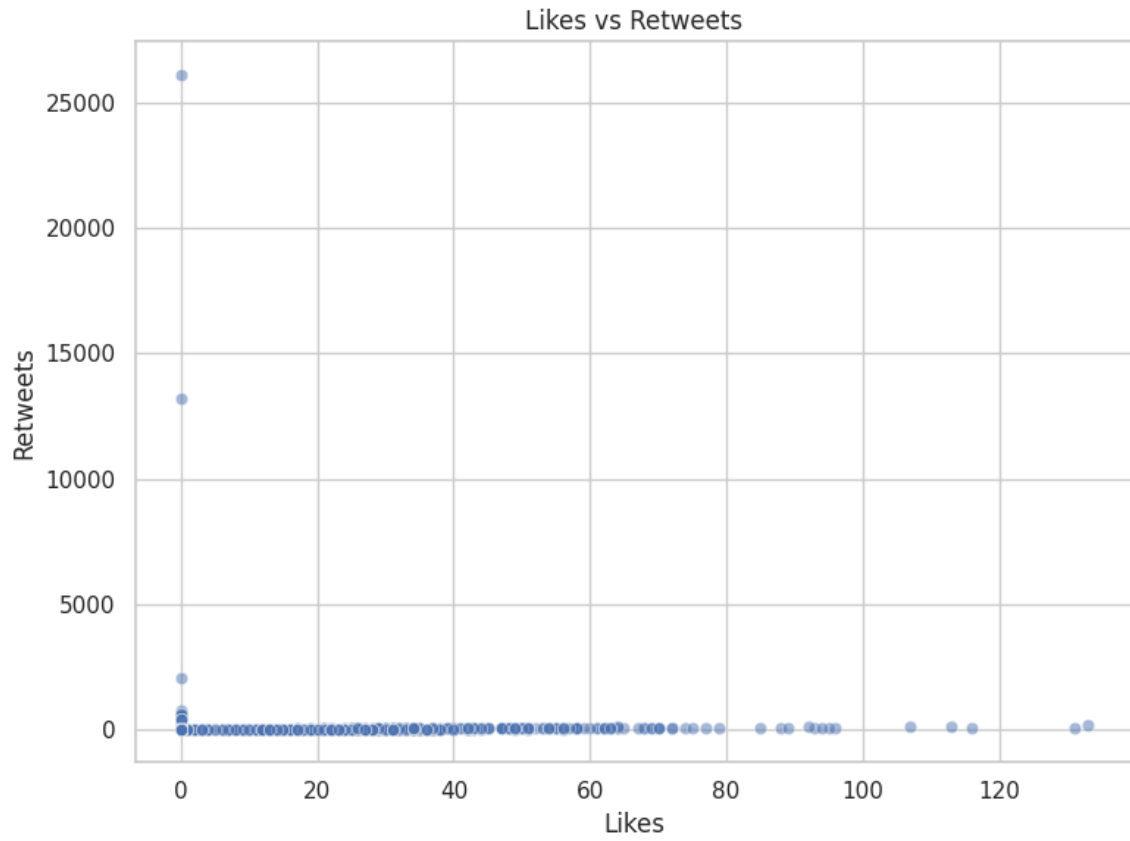
Put tweets_clean.csv in the same folder as your script/notebook.

Step 3: Run the Script

Visualization of outputs:







Conclusion:

This project successfully analyzes Twitter user timeline data and builds advanced predictive models for engagement forecasting. Using machine learning, time-series forecasting, and deep learning, the project identifies key behavioral patterns and determines which features contribute most to tweet engagement.

The visualizations offer clear insights into best posting times, influential users, and how different tweet-level features drive engagement. With an R^2 score of 0.852, the model performs well and can be integrated into future applications such as:

- Content recommendation systems
- Engagement prediction tools
- Social media dashboards
- Automated posting optimization systems